

Upstream of the Inflammasome

Dietary Neu5Gc, Ammonium Potentiators, and the Modest Effect of Colchicine on Coronary Plaque: A Testable Proposal Arising from the EKSTROM Trial

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STATUS OF THIS PAPER

This paper is a hypothesis-generating proposal, not a report of new data. It separates what is **established** in the published literature from what is **hypothesised** by the SGIC framework, and flags points that are **contested**. Each section's evidence tier is marked with a coloured callout.

Its practical purpose is to propose a low-cost secondary analysis of stored baseline serum from the EKSTROM colchicine trial, which would test the central claim directly.

Nothing in this paper is medical advice. Colchicine is a prescription drug with a narrow therapeutic margin and important drug interactions; it should not be started or stopped without a physician.

ABSTRACT

Low-dose colchicine reduces major adverse cardiovascular events in some settings (COLCOT, LoDoCo2) but not others (CLEAR SYNERGY), and in the EKSTROM imaging trial it reduced total percent atheroma volume over twelve months without significantly reducing low-attenuation plaque, the primary endpoint.¹⁻⁴ The effect is real but modest. Colchicine acts downstream: it impairs microtubule-dependent assembly of the NLRP3 inflammasome and neutrophil recruitment, blunting interleukin-1 β and interleukin-6 signalling regardless of what initiated the inflammation.

This paper proposes that, in a susceptible subset of people, one of the upstream initiators is antibody-mediated inflammation against dietary N-glycolylneuraminic acid (Neu5Gc) incorporated into vascular endothelium, a process termed xenosialitis. In human-like *Cmah*^{-/-}*Ldlr*^{-/-} mice, Neu5Gc loss increased atherosclerosis about 1.9-fold, and anti-Neu5Gc antibodies combined with a Neu5Gc-rich diet increased it further.⁵ If this pathway operates in humans, colchicine suppresses its output while the antigen continues to arrive. Removing the antigen, and the ammonium-derived potentiators that the SGIC framework proposes lower the inflammatory threshold, should therefore add to colchicine's effect rather than duplicate it.

The hypothesis makes a specific, falsifiable prediction: among EKSTROM participants, those with high baseline anti-Neu5Gc IgG, particularly against bovine-type Neu5Gc glycotopes, should show the largest colchicine-associated reduction in plaque. The trial's small size, serial coronary CT angiography, and probable stored serum make this an unusually cheap test. Counter-evidence, including the neutral-to-favourable dairy epidemiology and the methodological problems of anti-Neu5Gc assays, is set out alongside the argument.

1 Colchicine and the Residual Inflammatory Risk Problem

Lowering LDL cholesterol reduces cardiovascular events but leaves a residual risk that tracks with inflammatory markers. CANTOS established that targeting inflammation alone, by neutralising interleukin-1 β with canakinumab, reduces recurrent events without changing lipids.⁶ Colchicine is the inexpensive oral route to the same pathway.

1.1 Mechanism

Colchicine binds tubulin and disrupts microtubule polymerisation. In neutrophils and monocytes this impairs chemotaxis, adhesion and degranulation, and it interferes with the spatial assembly of the NLRP3 inflammasome, reducing caspase-1 activation and the release of IL-1 β and, downstream, IL-6 and C-reactive protein. In patients with acute coronary syndrome, colchicine acutely lowered transcoronary IL-1 β , IL-18 and IL-6.⁷ The drug therefore acts on the *response* to an inflammatory stimulus. It does not remove the stimulus.

1.2 The outcome trials

ESTABLISHED Colchicine's effect is consistent in direction but modest and setting-dependent

COLCOT (2019): colchicine 0.5 mg daily started within 30 days of myocardial infarction reduced the composite ischaemic endpoint by about 23%.²

LoDoCo2 (2020): in chronic coronary disease, colchicine 0.5 mg daily reduced the composite endpoint by about 31%.³

CLEAR SYNERGY / OASIS 9 (2025): in 7,062 patients started on colchicine immediately after PCI for acute MI and followed for a median of about three years, there was no reduction in the composite endpoint.⁴

1.3 The EKSTROM imaging trial

EKSTROM (Budoff and colleagues, 2026) is the first randomised, placebo-controlled trial to measure colchicine's effect on coronary plaque by serial coronary CT angiography.¹ Eighty-four patients with established stable coronary disease were randomised to colchicine 0.5 mg daily or placebo for twelve months, on top of standard therapy; 72 completed follow-up.

- **Primary endpoint (low-attenuation plaque volume):** no significant difference between groups.
- **Secondary endpoint (total percent atheroma volume):** median change +0.3 with colchicine versus +1.4 with placebo ($P = 0.008$ unadjusted; $P = 0.015$ in multivariable models).
- **Non-calcified and fibro-fatty plaque:** trends toward regression, not statistically significant.
- **Inflammatory markers:** reduced with colchicine, but not significantly.
- **Pericoronary adipose tissue attenuation:** unchanged in a substudy of 71 participants; mean baseline hsCRP was low, at 0.8 mg/L.⁸

In plain terms, colchicine slowed overall plaque growth in a modestly inflamed population without clearly shrinking the soft, lipid-rich plaque that is most prone to rupture. That is the pattern one would expect if colchicine is damping a process whose source remains active.

2 The Upstream Hypothesis: Xenosialitis of the Vascular Wall

2.1 Neu5Gc as a xenoautoantigen

ESTABLISHED Humans cannot make Neu5Gc, yet display it and make antibodies against it

Humans lost the *CMAH* gene about 2–3 million years ago and cannot synthesise Neu5Gc. Dietary Neu5Gc from red meat and dairy is nonetheless metabolically incorporated into human glycans, with preferential accumulation in tissues with high turnover, including endothelium and epithelia.^{9,10}

Anti-Neu5Gc antibodies are detected in most human sera, are polyclonal, and vary widely between individuals. Their induction in infancy coincides with the introduction of Neu5Gc-containing foods and gut colonisation, and is linked to Neu5Gc scavenged and displayed by non-typeable *Haemophilus influenzae*.^{10,11}

Human anti-Neu5Gc antibodies bind Neu5Gc-bearing human endothelial cells and can activate them.¹²

2.2 The proposed cascade in the arterial wall

The SGIC framework proposes the following sequence at the endothelium, which the author terms the Sialic-Glycan Inflammatory Cascade:

- **Antigen delivery.** Dietary Neu5Gc reaches the circulation and is incorporated into endothelial glycans, with the high shear, high turnover sites at arterial branch points plausibly receiving the most.
- **Antibody binding.** Circulating anti-Neu5Gc IgG recognises Neu5Gc in the context of its underlying glycan chain. The epitope configuration, not total Neu5Gc content, determines whether a given person's antibodies bind.
- **Sublytic complement attack.** Bound IgG fixes complement. Membrane attack complex inserted at sublytic levels does not kill the cell but triggers calcium flux and NLRP3 inflammasome activation, a link established in other cell types.¹³
- **Inflammasome output.** IL-1 β release recruits monocytes, promotes their conversion to lipid-laden macrophages, and drives the IL-6 and CRP signal that colchicine and canakinumab lower.

Colchicine intervenes at the last step. Dietary elimination intervenes at the first. The two do not compete; they act at opposite ends of the same chain.

HYPOTHESIS The central claim of this paper

In people whose antibody repertoire recognises the Neu5Gc configurations present in their diet, endothelial xenosialitis is a meaningful and continuous source of the residual inflammatory risk that colchicine partially suppresses.

It follows that eliminating the antigen should add to colchicine's effect on plaque, and that colchicine's benefit should be concentrated in this antibody-defined subgroup.

3 Supporting Evidence and the Case Against

3.1 The animal evidence

Kawanishi and colleagues, in the Varki and Gordts laboratories at UC San Diego, crossed human-like *Cmah*^{-/-} mice onto the atherosclerosis-prone *Ldlr*^{-/-} background.⁵ On a sialic-acid-free high-fat diet, Neu5Gc-deficient mice developed about 1.9-fold more atherosclerosis than controls with no change in lipoprotein profiles, driven by heightened macrophage cytokine output and hyperglycaemia; the authors termed this the intrinsic mechanism. When the mice were immunised to produce human-like anti-Neu5Gc antibodies and then fed a Neu5Gc-rich diet, atherosclerosis increased further, which the authors termed the extrinsic mechanism. A follow-up study found that dietary Neu5Ac competed with Neu5Gc for endothelial incorporation and protected against the atherosclerosis associated with Neu5Gc loss.¹⁴

CONTESTED How far the mouse model transfers to humans

Soulillou and colleagues argued that the model overstates the case: the mice need a strong, adjuvanted immunisation to produce anti-Neu5Gc antibodies, whereas humans acquire theirs through low-grade early-life exposure, and the *Ldlr*^{-/-} background has its own metabolic effects.¹⁵

The Varki group replied that animal extrapolation always has limits, but that the intrinsic hyperimmune and diabetic tendencies of the mice match the human condition, and that the xenoautoantigen–xenoautoantibody reaction is real in humans.¹⁶

A fair summary: the mouse data show that the pathway *can* accelerate atherosclerosis. They do not show that it *does* so, to a clinically important degree, in people.

3.2 Four objections this paper must answer

Objection 1 — The dairy epidemiology. Large prospective cohort meta-analyses find dairy intake neutral or modestly inversely associated with cardiovascular disease, and the association with red meat is modest and confounded by other mechanisms such as TMAO, haem iron and sodium.¹⁷ If dietary Neu5Gc drove atherosclerosis broadly, dairy should look harmful.

Response. The SGIC framework does not predict a population-wide effect. Under the three-gate model, only people whose antibody repertoire matches bovine-type Neu5Gc configurations (Gate One), who have barrier permeability admitting antigen with an adjuvant signal (Gate Two), and who lack sufficient local complement regulation (Gate Three) are susceptible. An effect confined to a minority is diluted to near-invisibility in population averages, and dietary cohort studies cannot see antibody status. This response is only persuasive if the subgroup can be identified in advance, which is what the proposal in Section 5 tests.

Objection 2 — The low inflammation in EKSTROM. Mean hsCRP of 0.8 mg/L means these patients had little systemic inflammation for colchicine to act on, which may be the whole explanation for the modest effect.

Response. This is a genuine alternative explanation and cannot be dismissed. However, hsCRP measures systemic spill-over. Antibody-mediated endothelial inflammation may be local and intermittent, tracking antigen exposure after meals, and could matter at the plaque level while barely registering in serum CRP. The two explanations make different predictions: if low inflammation is the whole story, baseline hsCRP should predict colchicine response and anti-Neu5Gc status should add nothing.

Objection 3 — The antibody assays are unreliable. A 2024 systematic review of 114 studies found that reported prevalence and titres of anti-Neu5Gc antibodies depend heavily on method, epitope choice, blocking agent and controls, so that trends between studies cannot be separated from experimental artefact.¹⁸ In particular, bovine serum albumin used as a blocking agent itself carries Neu5Gc and can absorb the antibodies being measured.

Response. Any test of this hypothesis must use a matched-pair sialoglycan microarray (Neu5Gc versus Neu5Ac glycans on identical scaffolds), with a non-mammalian blocking agent. That method exists, and the Varki and Padler-Karavani laboratories have used it at scale.^{19,20} Within a single trial analysed with one standardised assay, the between-study problem does not arise.

Objection 4 — CLEAR SYNERGY. If colchicine sometimes works and sometimes does not, the inflammatory pathway may simply be less important than hoped.

Response. The trials differ in timing (immediately after PCI versus weeks after MI or in stable disease), duration and population, and cardiologists continue to debate the discrepancy. The hypothesis here offers one additional contributor: if a meaningful part of the benefit is confined to an antibody-defined subgroup, trial results will vary with how many such patients each trial happens to enrol. This is speculative, but it is testable.

4 Ammonium Potentiators: A Separate Lever

A common simplification is to describe ammonium caramels as a source of Neu5Gc. They are not. Class III (E150c, ammonia caramel) and Class IV (E150d, sulphite ammonia caramel) colours are made by heating carbohydrates with ammonium compounds and contain no Neu5Gc. In the SGIC four-match framework they occupy a distinct role.

HYPOTHESIS The four-match model applied to the vascular wall

Neu5Gc is the xenoantigen: the target the antibodies bind.

Wheat amylase-trypsin inhibitors (ATIs) are the alarm signal: TLR4 agonists that prime innate immune cells.

Gliadins open the door: zonulin-mediated increases in intestinal permeability that increase antigen delivery.

Ammonium compounds lower the threshold: potentiators that make a given antigen load more likely to provoke a response.

These roles are not interchangeable. Removing Neu5Gc alone lowers the antigen load; removing the potentiators alone raises the threshold at which the remaining load provokes inflammation. The framework predicts that addressing both gives more than either alone.

The potentiator role of ammonium compounds is the least developed part of the framework and currently rests on the author's clinical observation and n-of-1 experience rather than on published mechanistic data. It is included here because it is part of the intervention the author proposes. It is not needed for the central claim, which can be tested with antibody data alone.

5 Testable Predictions and a Proposal to the EKSTROM Investigators

A hypothesis is only as useful as the result that would disprove it. Table 1 sets out what the SGIC framework predicts, and what result would count against it.

#	Prediction	Test	Result that would falsify it
P1	Colchicine's effect on plaque is larger in participants with high baseline anti-Neu5Gc IgG.	EKSTROM stored baseline serum; interaction of titre with treatment on change in PAV and LAP.	No interaction, or a larger effect in low-titre participants.
P2	Antibody specificity for bovine-type Neu5Gc glycotopes predicts response better than total titre.	Matched-pair sialoglycan microarray on the same samples.	Total titre predicts as well as or better than epitope specificity.
P3	Anti-Neu5Gc status adds predictive information beyond baseline hsCRP.	Multivariable model including both.	hsCRP explains the response and antibody status adds nothing.
P4	Eliminating dietary Neu5Gc lowers hsCRP and IL-6 in high-titre individuals, but not low-titre individuals.	Short controlled dietary crossover with inflammatory markers.	Equal or no change in both groups.

#	Prediction	Test	Result that would falsify it
P5	Elimination plus colchicine slows plaque progression more than colchicine alone in high-titre patients.	Randomised factorial trial with serial CCTA (only if P1–P4 are supportive).	No additional effect of elimination on plaque.

Table 1. Predictions arising from the hypothesis, in order of cost. P1–P3 require no new patients.

5.1 Why EKSTROM is the right first test

Large outcome trials such as LoDoCo2 and CLEAR SYNERGY enrolled thousands of patients, and a secondary antibody analysis would be a major undertaking. EKSTROM is different. It has 72 completers, each with quantified serial plaque measurements, which is precisely the continuous outcome needed to test an interaction. If baseline serum was banked, assaying it for anti-Neu5Gc IgG is a modest laboratory request.

The sample is small, and a negative result would not close the question. A positive interaction, however, would be a strong signal to justify P4 and P5. The request to the investigators is therefore narrow:

HYPOTHESIS The specific request

1. Is baseline serum or plasma from EKSTROM participants stored and available for secondary analysis under the existing consent?
2. If so, would the investigators consider a collaborative analysis of anti-Neu5Gc IgG, ideally by matched-pair sialoglycan microarray, against the change in total percent atheroma volume and low-attenuation plaque volume, stratified by treatment arm?
3. If the investigators prefer, the analysis can be pre-registered, with predictions P1–P3 stated in advance, so that the result counts regardless of which direction it goes.

6 Conclusion

Colchicine works on the response, not the cause. EKSTROM shows that its effect on coronary plaque is real but modest, and leaves open why. This paper offers one candidate explanation: that in an antibody-defined subgroup, dietary Neu5Gc incorporated into the arterial wall provides a continuous inflammatory stimulus that colchicine can damp but not remove.

The idea has animal support, credible objections and a clear test. It does not replace lipid lowering, and it does not justify anyone changing their medication. What it does justify is a look at serum that may already be sitting in a freezer, alongside plaque measurements that have already been made.

Declaration

The author is an independent researcher with no institutional affiliation, no funding for this work and no financial relationship with any manufacturer of colchicine or food products. The SGIC framework is the author's own work. Correspondence: hamerglycans.com.

References

1. Budoff MJ, Bhandari M, Iskander B, et al. Effect of colchicine on progression of known coronary atherosclerosis in patients with stable coronary artery disease: EKSTROM randomized placebo controlled trial. *Eur Heart J Cardiovasc Imaging* 2026;27(4):682–692. doi:10.1093/ehjci/jeag028
2. Tardif JC, Kouz S, Waters DD, et al. Efficacy and safety of low-dose colchicine after myocardial infarction (COLCOT). *N Engl J Med* 2019;381:2497–2505.
3. Nidorf SM, Fiolet ATL, Mosterd A, et al. Colchicine in patients with chronic coronary disease (LoDoCo2). *N Engl J Med* 2020;383:1838–1847.
4. Jolly SS, d'Entremont MA, Lee SF, et al. Colchicine in acute myocardial infarction (CLEAR SYNERGY / OASIS 9). *N Engl J Med* 2025;392:633–642. doi:10.1056/NEJMoa2405922
5. Kawanishi K, Dhar C, Do R, Varki N, Gordts PLSM, Varki A. Human species-specific loss of CMP-N-acetylneuraminic acid hydroxylase enhances atherosclerosis via intrinsic and extrinsic mechanisms. *Proc Natl Acad Sci USA* 2019;116(32):16036–16045. doi:10.1073/pnas.1902902116
6. Ridker PM, Everett BM, Thuren T, et al. Antiinflammatory therapy with canakinumab for atherosclerotic disease (CANTOS). *N Engl J Med* 2017;377:1119–1131.
7. Martínez GJ, Robertson S, Barraclough J, et al. Colchicine acutely suppresses local cardiac production of inflammatory cytokines in patients with an acute coronary syndrome. *J Am Heart Assoc* 2015;4:e002128.
8. Colasurdo JR, Mora R, Verghese D, et al. The effect of colchicine on pericoronary adipose tissue attenuation by serial coronary CT angiography: a substudy of the EKSTROM trial. *Am J Prev Cardiol* 2026;29:101572. doi:10.1016/j.ajpc.2026.101572
9. Tangvoranuntakul P, Gagneux P, Diaz S, et al. Human uptake and incorporation of an immunogenic nonhuman dietary sialic acid. *Proc Natl Acad Sci USA* 2003;100:12045–12050.
10. Dhar C, Sasmal A, Varki A. From “serum sickness” to “xenosialitis”: past, present, and future significance of the non-human sialic acid Neu5Gc. *Front Immunol* 2019;10:807.
11. Taylor RE, Gregg CJ, Padler-Karavani V, et al. Novel mechanism for the generation of human xeno-autoantibodies against the nonhuman sialic acid N-glycolylneuraminic acid. *J Exp Med* 2010;207:1637–1646.
12. Pham T, Gregg CJ, Karp F, et al. Evidence for a novel human-specific xeno-auto-antibody response against vascular endothelium. *Blood* 2009;114:5225–5235.
13. Laudisi F, Spreafico R, Evrard M, et al. Cutting edge: the NLRP3 inflammasome links complement-mediated inflammation and IL-1 β release. *J Immunol* 2013;191:1006–1010.
14. Kawanishi K, Coker JK, Grunddal KV, et al. Dietary Neu5Ac intervention protects against atherosclerosis associated with human-like Neu5Gc loss — brief report. *Arterioscler Thromb Vasc Biol* 2021;41(11):2730–2739.
15. Soulillou JP, Cozzi E, Galli C, Bach JM. Can we extrapolate from a Cmah $^{-/-}$ Ldlr $^{-/-}$ mouse model a susceptibility for atherosclerosis in humans? *Proc Natl Acad Sci USA* 2020;117:1845–1846.

16. Kawanishi K, Dhar C, Varki A, Gordts PLSM. Reply to Soulillou et al.: Difficulties in extrapolating from animal models exemplify unusual human atherosclerosis susceptibility and mechanisms via CMAH loss. *Proc Natl Acad Sci USA* 2020;117:1847–1848.
17. Guo J, Astrup A, Lovegrove JA, et al. Milk and dairy consumption and risk of cardiovascular diseases and all-cause mortality: dose-response meta-analysis of prospective cohort studies. *Eur J Epidemiol* 2017;32:269–287.
18. Hutton E, Scott E, Robson CN, Signoret N, Fascione MA. A systematic review reveals conflicting evidence for the prevalence of antibodies against the sialic acid ‘xenoautoantigen’ Neu5Gc in humans and the need for a standardised approach to quantification. *Front Mol Biosci* 2024;11:1390711.
19. Padler-Karavani V, Hurtado-Ziola N, Pu M, et al. Human xeno-autoantibodies against a non-human sialic acid serve as novel serum biomarkers and immunotherapeutics in cancer. *Cancer Res* 2011;71:3352–3363.
20. Bashir S, Fezeu LK, Leviatan Ben-Arye S, et al. Association between Neu5Gc carbohydrate and serum antibodies against it provides the molecular link to cancer: French NutriNet-Santé study. *BMC Med* 2020;18:262.